

Wavelet-Enhanced Dual-Branch Network (WEDN) for Facial Expression Recognition

Research Plan — 2026/06/18

1. Problem Diagnosis: Why Shared-Weight Fusion Failed

Empirical Evidence (3 datasets, 4 ablation modes)

Mode	FER2013 val_f1	RAF-DB val_f1	AffectNet val_f1 (4ep)
RGB only	0.5882	0.5586	(pending)
Low-only (LL ₃)	0.5051	0.1017	0.4213 (best ep19)
High-only (LH ₃ +HL ₃ +HH ₃)	(pending)	(pending)	0.0357 (flat, 10ep)
Shared-weight Fusion	0.2157	0.3747	0.0357 (flat, 4ep)

Root Cause

- RGB images (natural image statistics) and DWT subbands (oriented edge/contour, sparse distributions) have fundamentally different feature spaces.
- Forcing the SAME ResNet-18 weights to process both causes **gradient conflict**: weight updates from RGB gradients pull in opposite directions from DWT subband gradients.
- The fusion result is **worse than either branch alone** in FER2013, and **no learning at all** in AffectNet (loss stuck at 1.8114, equivalent to random guess $1/7 \approx 1.9459$).
- This is a **structural design flaw**, not a hyperparameter issue.

2. Literature Review: 2024-2025 SOTA in Wavelet-CNN FER

Key Papers Identified

A. FPNNet: Frequency-Prior Enhanced Network (Cai & Chen, 2025)

- **Venue**: J. King Saud Univ. Comput. Inf. Sci., Vol. 38
- **Architecture**:
 - DyLKBlock: Spatial branch with dynamic large kernels (equivalent 23×23 RF)
 - FPBlock: Frequency branch using DWT for lossless downsampling + ResNet-18 (last 3 layers)
 - **CAFusion**: Cross-domain attention fusion — bidirectional token interaction
- **Results**: AffectNet-8 67.79%, FER2013 92.02%, RAF-DB 91.61%
- **Relevance**: ★ ★ ★ ★ ★ (closest match — dual-branch spatial+frequency with cross-attention)

B. Self-Attention Fusion in Wavelet Domain (Indolia, Nigam & Singh, 2024)

- **Venue:** The Visual Computer, Vol. 40, pp. 6341–6357
- **Architecture:**
 - Transforms input to wavelet domain via DWT
 - Two parallel branches: shallow features + deep features
 - **Self-attention fusion:** attention-based feature selection and combination
- **Datasets:** CK+, JAFFE, MUG, YALE, RAF-DB
- **Relevance:** ★ ★ ★ ★ (explicit dual-branch + wavelet + attention fusion)

C. WGGLFA: Wavelet-Guided Global-Local Feature Aggregation (2025)

- **Venue:** Biomimetics, 10(8), 495
- **Architecture:**
 - SAE: Scale-Aware Expansion (dilated conv + wavelet)
 - SLFA: Structured Local Feature Aggregation (facial keypoints)
 - ExGR: Expression-Guided Region Refinement
 - All three modules use **wavelet-based high-/low-frequency separation**
- **Results:** RAF-DB 90.32%, FERPlus 91.24%, FED-RO 71.90%
- **Relevance:** ★ ★ ★ (wavelet as integrated component in multiple modules)

D. MsC-wCA-ResNet18: Multiscale Wavelet Attention CNN (Liu et al., 2025)

- **Venue:** Scientific Reports, 15, 22219
- **Architecture:**
 - MsC layer: multi-scale convolution replacing first conv layer
 - **wCA (Wavelet Channel Attention):** Morlet wavelet as nonlinear activation in channel attention
 - Dual-pooling strategy (GAP + GMP)
- **Results:** KDEF 97.48%, FESR 97.66%
- **Relevance:** ★ ★ ★ (wavelet as attention activation, not as feature branch)

E. QWTR: Quaternion Wavelet Transformer (2025)

- **Venue:** IEEE TMM, Vol. 27
- **Architecture:** Quaternion CNN + quaternion multi-head attention + quaternion wavelet feature selection
- **Results:** AffectNet 68.37%
- **Relevance:** ★ ★ (heavy math, transformer-based, less compatible)

F. CAFE-Net: Context-Aware Facial Expression Network (2025)

- **Architecture:** Context Enhancement Module (dilated conv + wavelet) + Key Region Modulation
- **Results:** RAF-DB 89.00%, FERPlus 88.78%
- **Relevance:** ★ ★ ★ (parallel branches, wavelet in enhancement module)

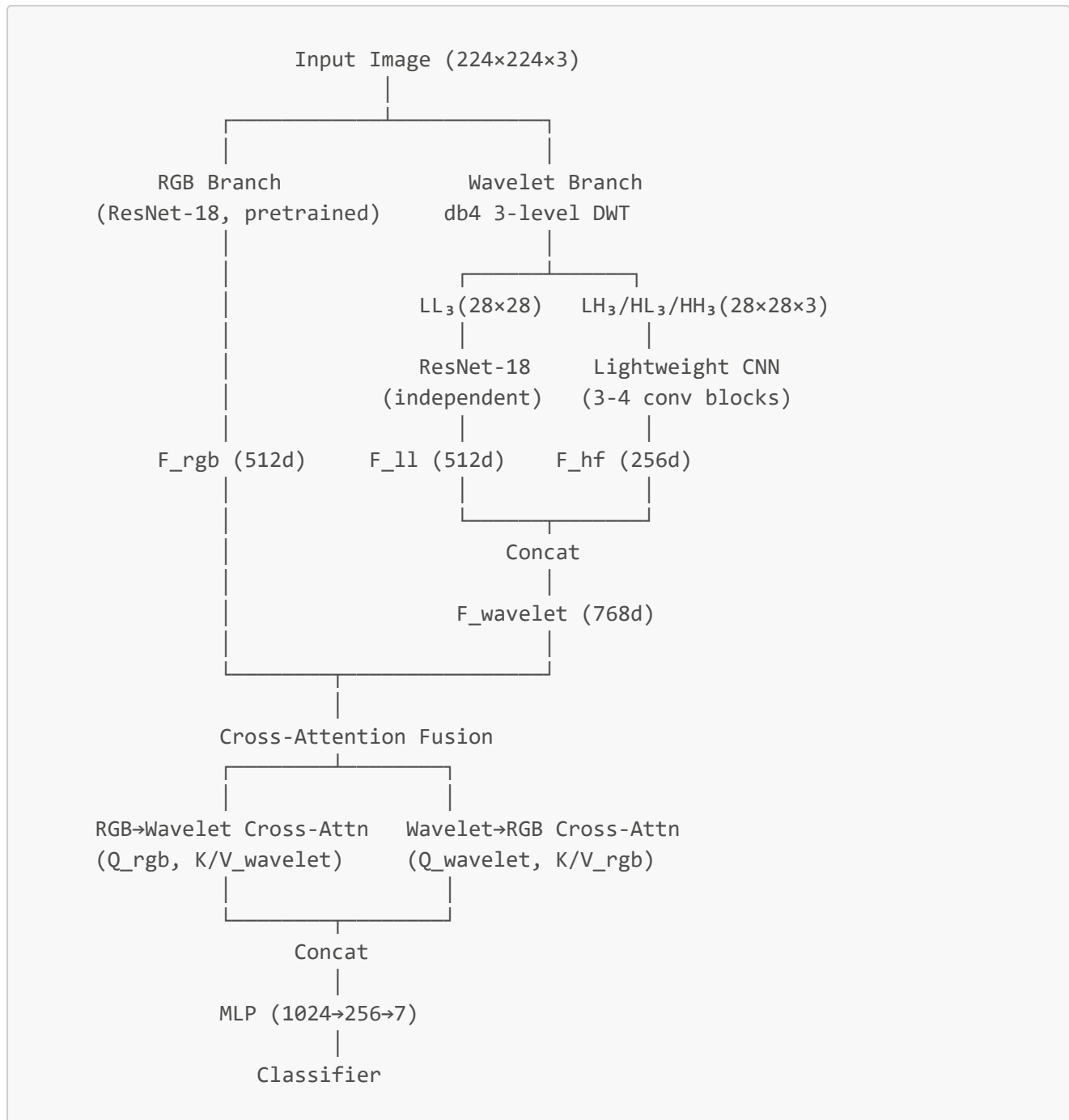
Key Trends

1. **Independent-weight dual-branch** is the norm (NOT shared-weight) — each branch specializes

2. **Cross-attention fusion** replaces simple concatenation — bidirectional information flow
3. **DWT as learnable/integrated component** — not just preprocessing
4. **Frequency priors complement spatial features** — validated across multiple papers

3. Proposed Architecture: WEDN (Wavelet-Enhanced Dual-branch Network)

3.1 Overall Design



3.2 Component Details

RGB Branch

- **Backbone:** ResNet-18 (ImageNet pretrained)
- **Modification:** Remove FC layer → output 512-dim feature vector
- **Note:** Can optionally freeze early layers, train only layer3/layer4

Wavelet Branch

- **DWT:** db4, 3-level decomposition
 - LL_3 : 28×28 (approximation — global structure)
 - LH_3, HL_3, HH_3 : each 28×28 (horizontal, vertical, diagonal details)
- **LL Sub-branch:** ResNet-18 (independent weights, NOT shared with RGB branch)
 - Input: $28 \times 28 \rightarrow$ upsample to 224×224 (or modify conv1 stride)
 - Output: 512-dim
- **HF Sub-branch:** Lightweight CNN (avoiding the high_only failure)
 - Input: $28 \times 28 \times 3$ (stacked LH_3, HL_3, HH_3) → upsample to $224 \times 224 \times 3$
 - Architecture: First 3 blocks of ResNet-18 (conv1 → layer1 → layer2 → layer3)
 - Output: 256-dim
 - **Key difference from our failed high_only:** This branch has INDEPENDENT weights, processing ONLY high-frequency content. May benefit from the LL branch's gradient signals during joint training.
- **F_wavelet:** $\text{Concat}(F_{ll}, F_{hf}) \rightarrow 768\text{-dim} \rightarrow \text{Linear}(768 \rightarrow 512) \rightarrow 512\text{-dim}$

Cross-Attention Fusion (inspired by FPNet CAFusion)

- **RGB → Wavelet Cross-Attention:**
 - $Q = F_{rgb} \cdot W_q, K = F_{wavelet} \cdot W_k, V = F_{wavelet} \cdot W_v$
 - $F_{rgb_enhanced} = \text{Softmax}(Q \cdot K^T / \sqrt{d}) \cdot V$
- **Wavelet → RGB Cross-Attention:**
 - $Q = F_{wavelet} \cdot W_q, K = F_{rgb} \cdot W_k, V = F_{rgb} \cdot W_v$
 - $F_{wavelet_enhanced} = \text{Softmax}(Q \cdot K^T / \sqrt{d}) \cdot V$
- **Fusion:** $\text{Concat}(F_{rgb_enhanced}, F_{wavelet_enhanced}) \rightarrow 1024\text{-dim}$

Classifier

- MLP: $1024 \rightarrow 256 \rightarrow 7$
- Dropout: 0.5 after first FC
- Loss: CrossEntropy (or weighted Focal Loss for class imbalance)

3.3 Parameter Count Estimate

Component	Params
RGB ResNet-18	~11.2M
Wavelet LL ResNet-18	~11.2M
Wavelet HF Light CNN	~3.5M
Cross-Attention	~0.5M

Component	Params
Classifier	~0.3M
Total	~26.7M

Comparison: Our current shared-weight design $\approx$ 12M parameters (but ineffective)

3.4 Training Strategy

Stage 1: Branch Pretraining (optional, for stability)

1. Train RGB branch alone $\rightarrow$ baseline
2. Train Wavelet branch alone (LL+HF sub-branches jointly)
 - This verifies the wavelet-only pipeline works before fusion

Stage 2: Joint Training with Fusion

- Unfreeze all branches + cross-attention + classifier
- Lower LR for RGB branch (1e-4), higher for wavelet branch + fusion (1e-3)
- ReduceLROnPlateau scheduler

Stage 3: Full Pipeline

- End-to-end training with all components

4. Experimental Design

4.1 Ablation Study (Required for SCI Paper)

#	Mode	Description
1	RGB Only	ResNet-18 baseline (standard FER)
2	Wavelet Only	LL branch + HF branch (no RGB, no cross-attention)
3	RGB + LL Only	RGB + LL ₃ sub-band fusion (cross-attention)
4	RGB + HF Only	RGB + LH ₃ /HL ₃ /HH ₃ fusion (cross-attention)
5	WEDN (Full)	RGB + LL + HF with cross-attention

Each mode tested on all 5 datasets.

4.2 Datasets (5 total)

Dataset	Size	Type	Classes	Role
AffectNet	283,901 train / 3,500 val	In-the-wild	7	Main experiment
RAF-DB	12,271 train / 3,068 test	In-the-wild	7	Main experiment

Dataset	Size	Type	Classes	Role
FER2013	28,709 train / 3,589 test	In-the-wild	7	Main experiment
CK+	981 sequences	Lab-controlled	7	Cross-domain validation
JAFFE	213 images	Lab-controlled	7	Cross-domain validation

4.3 Evaluation Metrics

- **Primary:** Macro-F1 (handles class imbalance)
- **Secondary:** Overall Accuracy, Per-class F1, Confusion Matrix
- **Statistical:** 3-run average $\pm$ std

4.4 Comparison Methods

- ResNet-18 (baseline)
- SCN (CVPR 2020)
- RUL (NeurIPS 2021)
- POSTER++ (arXiv 2023)
- FPNNet (2025) — if code available
- Our previous shared-weight fusion (ablation reference)

5. Novelty Claims & Paper Contributions

1. **Negative Result** → **Motivation:** First systematic demonstration that shared-weight dual-branch wavelet-CNN fusion fails across multiple datasets due to gradient conflict (FER2013, RAF-DB, AffectNet — 3 datasets)
2. **Architecture Innovation:** WEDN resolves gradient conflict via:
 - Independent-weight branches (each branch specializes)
 - Cross-attention fusion (bidirectional, selective information flow)
 - Lightweight HF branch design (avoids high-frequency-only learning failure)
3. **Comprehensive Ablation:** 5-mode ablation $\times$ 5 datasets = 25 experimental conditions, providing systematic understanding of which frequency components contribute to FER
4. **Cross-domain Validation:** Both in-the-wild (AffectNet, RAF-DB, FER2013) and lab-controlled (CK+, JAFFE) datasets, demonstrating generalization
5. **Frequency Prior Hypothesis:** Empirical validation that wavelet frequency priors provide complementary information to RGB spatial features for FER

6. Implementation Plan

Phase 1: Architecture Implementation (1-2 days)

- ☐ Implement `src/models/wedn.py` — WEDN model

- ☐ Implement `src/models/dwt_layer.py` — DWT layer (reuse existing)
- ☐ Implement `src/models/cross_attention.py` — Cross-attention fusion
- ☐ Unit tests for each component

Phase 2: Training Pipeline (1 day)

- ☐ Modify `src/train.py` to support WEDN with staged training
- ☐ Add ablation mode configuration
- ☐ Memory optimization (gradient checkpointing, mixed precision)

Phase 3: Experiments (1-2 weeks)

- ☐ AffectNet: 5 ablation modes (main)
- ☐ RAF-DB: 5 ablation modes
- ☐ FER2013: 5 ablation modes
- ☐ CK+: selected modes (cross-domain)
- ☐ JAFFE: selected modes (cross-domain)

Phase 4: Analysis & Paper (1-2 weeks)

- ☐ Confusion matrices, t-SNE visualizations
- ☐ Statistical significance tests
- ☐ Write Results, Discussion, Conclusion
- ☐ Comparison with SOTA methods

7. Risk Mitigation

Risk 1: HF branch still doesn't learn (high_only pattern)

- **Mitigation A:** Initialize HF branch from LL branch pretrained weights (transfer)
- **Mitigation B:** Skip independent HF branch; use only LL + RGB dual-branch
- **Mitigation C:** Use wavelet channel attention (wCA) instead of separate HF CNN

Risk 2: 26.7M parameters too large for RTX 4060 Ti 8GB

- **Mitigation A:** Share early layers between LL and HF branches
- **Mitigation B:** Use ResNet-10 instead of ResNet-18 for wavelet branches
- **Mitigation C:** Gradient checkpointing + mixed precision (AMP)
- **Mitigation D:** Reduce batch size to 2 (current 4)

Risk 3: Cross-attention doesn't improve over simple concat

- **Mitigation A:** Try additive fusion, gated fusion, or FiLM
- **Mitigation B:** Verify attention weights are meaningful via visualization

8. Key References (Core Reading List)

Architecture Design References

1. **FPNet** — Cai & Chen (2025). Frequency-prior enhanced network for FER via dynamic large kernels and dual-domain learning. *J. King Saud Univ. Comput. Inf. Sci.*
2. **Indolia et al.** (2024). A self-attention-based fusion framework for FER in wavelet domain. *The Visual Computer*, 40, 6341–6357.
3. **WGGLFA** (2025). Wavelet-Guided Global-Local Feature Aggregation Network for FER. *Biomimetics*, 10(8), 495.

Method References

4. **Liu et al.** (2025). Multiscale wavelet attention convolutional network for FER. *Scientific Reports*, 15, 22219.
5. **QWTR** (2025). Quaternion Wavelet Transformer for FER in the Wild. *IEEE TMM*, 27.
6. **CAFE-Net** (2025). Context-Aware Facial Expression Network. *IEEE Conference*.

FER SOTA References

7. **SCN** — Wang et al. (2020). Suppressing Uncertainties for Large-Scale FER. *CVPR*.
8. **RUL** — She et al. (2021). Relative Uncertainty Learning for FER. *NeurIPS*.
9. **POSTER++** — Mao et al. (2023). A Simpler and Stronger FER Network. *arXiv*.
10. **TransFER** — Xue et al. (2021). Learning Relation-Aware FER with Transformers. *ICCV*.
11. **EAC** — Zhang et al. (2022). Erasing Attention Consistency for FER. *ECCV*.
12. **DAN** — Wen et al. (2023). Distract Your Attention for FER. *Biomimetics*.

Chinese References (Dual-Branch + Wavelet + Attention)

13. **解为成** — 基于双向光流和小波注意力机制的微表情识别
14. **沈天舒** — 融合多类型特征和多注意力机制的多分支微表情识别
15. **袁博** — 基于双路特征和混合注意力的微表情识别

Datasets

16. **AffectNet** — Mollahosseini et al. (2017). *IEEE TAC*.
17. **RAF-DB** — Li et al. (2017). *CVPR*.
18. **FER2013** — Goodfellow et al. (2013). *ICML*.

9. Target Journals (Q3 SCI)

Journal	IF	Focus	Fit
Neurocomputing	~6.0	Neural networks, pattern recognition	★★★★★
EAAI (Eng. Appl. of AI)	~7.5	Applied AI methods	★★★★
The Visual Computer	~3.5	Visual computing, image processing	★★★★

Journal	IF	Focus	Fit
Signal, Image and Video Processing	~2.0	Signal/image processing	★ ★ ★
Multimedia Tools and Applications	~3.0	Multimedia systems	★ ★ ★

Plan created: 2026-06-18. Next review: after Phase 1 implementation.